

OpenCaseLaw: An Open Multilingual Legal Knowledge Graph for Switzerland

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Abstract

We present **OpenCaseLaw**, an open, continuously-refreshed corpus of **971,992** Swiss court decisions across **108 courts** spanning **26 Swiss cantons, the federal jurisdiction, and the Swiss-relevant subset of the European Court of Human Rights** (28 jurisdictional layers), in German, French, and Italian, paired with a citation graph of **8.65M** distinct outgoing references (**8.09M** resolved, 93.5 % coverage) and a statute graph of **11.26M** edges over **283,119** distinct provisions. The corpus is bridged to **5,516** federal and **15,722** cantonal laws (**753,842** articles in total), **362** CC-BY scholarly commentaries, and **5,292** Federal Council Botschaften (Materialien, DE/FR/IT-parallel, with **381,711** full-text paragraphs and **8,124** article-anchored links). The infrastructure is served through a deployed MCP/REST endpoint at `mcp.opencaselaw.ch` (33 tools), a Word-add-in client, and a public dashboard. Snapshot: 2026-05-13. We use the resource to introduce three contributions: (i) a released corpus and knowledge graph with documented provenance, licensing, and a four-layer dataset-health framework (63 codified checks, drift detection, and a publish-gate that blocks production updates on critical regression); (ii) a *regeste-derived cross-lingual retrieval diagnostic* with parallel keyword queries in DE/FR/IT against 50 highly-cited decisions. Because each query is extracted from its target case’s own multilingual regeste, the absolute scores are an upper bound on realistic legal-research retrieval; we therefore report per-direction asymmetries as the interpretable signal (Sec. 7, Sec. 11); (iii) a deployed five-rail closing audit that checks named legal-RAG error classes deterministically (case-citation existence, statute resolution, verbatim-quote source matching, decision-date sanity) plus an opt-in proposition-grounding judge, with per-class metrics and explicit calibration limits (Sec. 6, Sec. 11). We do *not* claim a general hallucination-rate reduction. Released artifacts (corpus packaging, citation/statute graphs, benchmark) follow record-level upstream terms: code is MIT; original-source decisions are official acts of Swiss authorities and inherit their public-domain status; commentaries retain CC-BY-4.0 / CC-BY-SA-4.0; the `entscheide.suche.ch` mirror is CC-BY-4.0. A public dashboard tracks live dataset health.

1 Introduction

Legal artificial intelligence has converged on retrieval-augmented generation (RAG) as the standard architecture for question answering over jurisprudence. Yet recent measurement studies report fabrication rates of 58–88 % on general-purpose LLMs (ChatGPT-4 to Llama-2 range) [3] and 17–33 % on commercial legal-RAG products [7]. The mitigation literature is methodologically sophisticated but concentrated on *English-language, single-jurisdiction* settings where retrieval substrates and ground-truth citation graphs are limited.

Switzerland is the most multilingual major civil-law jurisdiction. The authoritative version of a leading Swiss federal decision may be in German, French, or Italian, and citations cross

language boundaries: in our resolved citation graph, **34.0 % of resolved citations are cross-lingual** (Table 3). Building a credible RAG system for Swiss law therefore requires (a) a corpus that spans the federal-cantonal landscape across all three languages, (b) a citation and statute graph that can verify generated claims, and (c) evaluation that measures cross-lingual retrieval as a first-class concern.

We present **OpenCaseLaw**, an open infrastructure that fills these gaps. We frame this work as a *resource paper with diagnostic evaluations*: the corpus, citation graph, and statute graph are the contribution; the cross-lingual numbers and the audit pipeline are diagnostics that characterise specific failure modes, not general claims about retrieval quality or hallucination mitigation.

C1 — Released corpus and knowledge graph. We release an open Swiss legal knowledge graph built from a continuously refreshed corpus of court decisions, resolved case-to-case citations, statute references, legislative materials (Federal Council Botschaften), and open commentaries. The resource covers federal, cantonal, and Swiss-relevant ECHR decisions in German, French, and Italian. We document provenance, licensing per record category, and operate a four-layer dataset-health framework that codifies 63 invariants with drift detection and a publish-gate that blocks production updates on critical regression (Section 9).

C2 — Regeste-derived cross-lingual retrieval diagnostic. We provide a parallel-query set in DE/FR/IT against 50 highly-cited Swiss decisions, with each query extracted from the target case’s own official regeste. The resulting $\text{MRR@10}=0.630$ is an *upper bound* on realistic retrieval performance because the query and target share construction vocabulary; we therefore foreground the directional asymmetries ($\text{IT}\rightarrow\text{DE}=0.776$ vs. $\text{DE}\rightarrow\text{FR}=0.333$) rather than the absolute headline (Section 7, Section 11).

C3 — Deterministic legal-RAG audit layer. We deploy a five-rail closing audit (Section 6) that checks specific legal-RAG error classes: case-citation existence (R1), statute resolution (R2), verbatim-quote source matching (R3), decision-date sanity (R4), and proposition grounding (R5, judge-mediated, advisory). We report per-rail metrics on the 30-question Swiss Legal RAG bench under a prior-only condition (R1 unresolvable-citation rate, R5 false-positive rate of 41.7 %), and we do *not* claim a general hallucination-rate reduction.

2 Related work

Open legal corpora. The Caselaw Access Project [5] releases 6.7M US decisions but does not publish a citation graph or maintain a retrieval system. The *Open Legal Data* platform [11] releases German court decisions with a citation network but no statute graph and no multi-lingual coverage; *LeCNet* [4] introduces an Indian-judicial-context citation-network benchmark (26 k case-judgment nodes) for missing-citation recommendation. *Pile of Law* [6] and *MultiLegalPile* [9] are pretraining corpora rather than retrieval substrates. Within Swiss legal NLP, *Swiss-Judgment-Prediction* [8] is a binary outcome-prediction task; the *One Law, Many Languages* benchmark [14] bundles diverse Swiss-legal NLP tasks (document-to-document IR, court-view generation, leading-decision summarisation, citation extraction, classification) across five languages; *Swiss Landmark Decisions Summarisation* [13] is a 20 k-decision dataset where each ruling is paired with headnotes in all three official languages (DE/FR/IT) for cross-lingual summarisation. All three target narrow downstream NLP tasks rather than the underlying retrieval substrate; we release the corpus + citation graph + statute graph + Materialien layer that such tasks can be built on top of. The closest Swiss-specific comparator is the community project *entscheidsuche.ch* [10], which aggregates raw HTML/PDF feeds from cantonal and federal court portals into a unified harvesting layer (CC BY 4.0). *entscheidsuche.ch* is in fact one of

Table 1: OpenCaseLaw corpus overview. Snapshot 2026-05-13.

Quantity	Value
Total decisions	971,992
Distinct courts	108
Cantons + federal	28
Date range	1875–2026
German (de)	449,575 (46.3%)
French (fr)	441,158 (45.4%)
Italian (it)	80,704 (8.3%)

our upstream sources — many of our cantonal scrapers federate via its JSON feeds. Critically, however, *entscheidsuche.ch* provides no citation graph, no statute mirror, no Materialien layer, no retrieval API, and no cross-court normalisation, dedup, or multilingual joins; we build all of these on top.

Legal-RAG benchmarks. *LegalBench-RAG* [12] introduces retrieval evaluation for US legal RAG with $\sim 79\text{M}$ characters of human-annotated source material. *Butler & Butler 2026* [2] adds end-to-end correctness $\times$ groundedness $\times$ retrieval-accuracy decomposition on Victorian criminal law. *Bar Exam QA / Housing Statute QA* [16] are US-focused legal-research QA benchmarks. None of these benchmarks measure cross-lingual retrieval where the target authoritative version may be in a different language than the query.

Hallucination measurement and mitigation. *Dahl et al. 2024* [3] and *Magesh et al. 2024* [7] establish hallucination rates on general LLMs and commercial legal-RAG. *Self-RAG* [1] embeds critique directly in the generator via model-emitted reflection tokens, controlling retrieval and self-evaluation inline. *CRAG* [15] adds a lightweight retrieval evaluator that scores retrieved documents and triggers corrective web search when confidence is low. We instead decompose verification into per-error-class deterministic rails (case-citation existence, statute resolution, verbatim-quote source matching, decision-date sanity) plus an opt-in proposition-grounding judge.

Positioning. To our knowledge, OpenCaseLaw is the first open Swiss legal resource that bundles cross-court case-law aggregation, resolved case-to-case citation links, statute references, legislative-material (Botschaft) links, scholarly commentary anchors, and a public retrieval API in a single release. Each component individually has prior art (cited above); the contribution is the integrated graph, the multilingual scope, and the operational health framework around it. The cross-lingual diagnostic and the audit layer accompany the resource but are not the headline claim.

3 Resource: corpus

The OpenCaseLaw corpus comprises **971,992 decisions** from **108 distinct courts** across **26 Swiss cantons plus federal and ECHR jurisdictions** (28 jurisdictional layers). The federal Supreme Court (BGer) is refreshed hourly (Mon–Fri 05:00–16:00 UTC) via a dedicated Neuheiten poller; all other federations (BVGer, BStGer, BPatGer, regulatory authorities, cantonal portals, ECHR-Swiss-subset, historical archives) are refreshed by a nightly publish at 03:00 UTC. Coverage spans **1875–2026** and three official languages: German (449,575 / 46.3%), French (441,158 / 45.4%), Italian (80,704 / 8.3%). The federation aggregates source-specific

Table 2: Citation graph. Edges extracted by full-text scanning every decision in the corpus; resolved means the cited target is identified in the corpus. Snapshot 2026-05-13.

Quantity	Value
Decisions analysed for outgoing citations	971,852
Citation edges (source $\rightarrow$ target reference)	8,649,879
Edges with target resolved to a corpus decision	8,089,112 (93.5%)
Distinct decisions cited at least once	261,278
Decisions cited ≥ 100 times	10,874
Decisions cited $\geq 1,000$ times	999
Decisions cited $\geq 10,000$ times	47

Table 3: Cross-lingual citation matrix. Rows are source decision languages; columns are cited (target) decision languages. Cells give resolved citation edges. Off-diagonal = 2,749,064 / 8,087,818 = 34.0% of resolved citations with identified source and target language.

	$\rightarrow$ DE	$\rightarrow$ FR	$\rightarrow$ IT
DE $\rightarrow$	2,772,516	458,531	31,886
FR $\rightarrow$	1,937,750	2,519,278	63,429
IT $\rightarrow$	168,256	89,212	46,960

scrapers and adapters ranging from federal courts (BGer, BVGer, BStGer, BPatGer) to per-canton portals to the European Court of Human Rights (Swiss subset; see below), historical archives (BGE 1875–1953, MKG 1915–2025), and regulatory authorities (FINMA, WEKO, EDÖB, UBI, ELCom, PostCom, ComCom).

ECHR Swiss subset. The Swiss-relevant subset of European Court of Human Rights jurisprudence (**2,788 decisions**: 1,153 chamber + 93 grand-chamber + 230 committee + 835 HUDOC entries + 477 BGE-EGMR mirror) is integrated as a first-class federation. The set is refreshed daily; on the snapshot date the most recent ingest carried a same-day Article 11 violation against Switzerland (application 30781/22, decided 2026-05-07). The subset matters because **Article 6 ECHR** is the second-most-cited statutory provision in our graph (after federal-court procedure), and the official versions of ECHR-Switzerland judgments are typically French and English, requiring a multilingual retrieval substrate.

Completeness. Some source portals impose access restrictions (geographic, rate, or contractual). We document affected sources in the per-source coverage report, do not claim full coverage where access is restricted, and use only legally authorized collection methods. For closed historical archives where contemporaneous collection is no longer possible, we recover from publicly archived snapshots (e.g. Wayback Machine; ~ 903 BL ALS judgments).

Licensing. Court decisions in Switzerland are official published acts; the corpus packaging and added metadata are released CC0 to the extent rights exist. Upstream sources retain their own terms: scholarly commentaries are CC-BY-4.0 (OnlineKommentar.ch) and CC-BY-SA-4.0 (OpenLegalCommentary.ch); the cantonal-level mirror entscheidensuche.ch [10] which we federate as one input source is CC-BY-4.0. Decision IDs are stable across daily refreshes barring documented corrections (see §9 case studies).

Table 4: Citation-resolution precision proxies (computed over all 8,097,538 resolved pairs). Date-sanity pass = % of pairs where target decision date precedes the source’s; violations are logical impossibilities and thus a hard floor on false-positive count. Self-cite = pairs where source = target. Conf. p50 = median resolver confidence per stratum. Full per-stratum manual precision adjudication is deferred to v2.0 (see `benchmarks/citation_precision_audit.py`).

match_type	n	date sanity	self-cite	conf p50
docket_norm	3,286,234	97.25%	0	0.99
bge_bare	2,943,028	99.75%	0	0.85
bge_norm	1,215,768	99.26%	0	0.75
bge_pincite	652,508	97.35%	0	0.75
Overall	8,097,538	98.47%	0	—

4 Resource: knowledge graph

Citation graph. We extract citation edges by full-text scanning every decision in the corpus (971,852 decisions in the reference-graph snapshot; the FTS5 search index carries 140 more decisions ingested between graph rebuild and FTS5 swap), recovering **8,649,879 distinct outgoing references** of which **8,089,112 (93.5 %)** resolve to a target decision in the corpus. Resolution proceeds in three exact-match passes (general docket-normalisation; BGE/ATF/DTF prefix-form; bare BGE numeric form), augmented by a pin-cite-aware fallback for Swiss legal citations such as *BGE 125 V 352* that pinpoint into the body of a case starting at *BGE 125 V 351*. The pin-cite resolver matches each unresolved BGE-form reference to the nearest preceding first-page in the same volume + division within 30 pages, contributing **652,310** additional matches at a -0.10 confidence discount versus exact matches. The remaining 560,767 unresolved pairs (6.5 %) are dominated by cantonal or lower-court references genuinely outside our current coverage; the residual is BGE volumes we do not hold and malformed extraction artefacts (see `benchmarks/citation_resolution_analysis.py`). The deployed 93.5 % resolution is an 18.3pp lift over the prior release’s 75.2 %, recovered by fixing two implementation bugs: docket-normalisation drift in 2007–2009 *Bundesgericht* dockets stored with a space (8C 862_2008) versus the underscore form (8C_862_2008) emitted by the citation extractor, and the pin-cite mismatch above.

The citation graph is heavy-tailed (Table 2): 47 decisions are cited 10,000+ times, 952 are cited 1,000–9,999 times, and the rest follow a power law. The single most-cited decision is BGE 125 V 351 (85,108 incoming citations), the foundational case on the evidentiary weight of party-commissioned medical reports in social-insurance litigation.

Precision proxies. The 93.5 % resolution number above is a *coverage* metric — the share of extracted citation tokens that match a corpus decision — and does not by itself bound precision. A per-stratum manual precision audit (400 stratified samples, over-sampling the pin-cite fallback; harness shipped at `benchmarks/citation_precision_audit.py`) is deferred to a multi-annotator v2.0 effort. In its place we report three automated proxies (Table 4) that bound the worst case: every counted violation is a guaranteed false positive. *Date sanity* (target decision must be decided before source) passes 98.47 % over all 8.10M resolved pairs with both dates known — a hard lower bound on overall precision. The pin-cite stratum scores lowest (97.35 %), consistent with its heuristic nature; the exact-match strata (`bge_bare`, `bge_norm`) sit above 99.2 %. *Self-citation* is zero across the full graph. *Confidence percentiles* per match_type are reported alongside as a descriptive signal. These proxies are necessary but not sufficient; full precision intervals from human adjudication remain a v2.0 deliverable.

Cross-lingual citation. Table 3 reports the language-by-language citation matrix. **34.0 % of resolved citations with identified source and target language are cross-lingual:** French decisions cite German decisions **1.94M** times (FR→DE dominates the off-diagonal), German decisions cite French decisions **0.46M** times, and Italian decisions cite German decisions **0.17M** times. The asymmetry — DE↔FR is bidirectional but FR→DE outweighs DE→FR by 4.2×, while IT→DE/FR (0.24M combined) dominates IT→IT (46,960) by ~5× — reflects practitioners’ practical reliance on German jurisprudence as binding precedent in mixed-language litigation.

Statute graph. We resolve **11,261,717 decision-to-statute edges** touching **283,119 distinct provisions**. The statute layer is grounded in two parallel mirrors: a federal layer of **5,516 SR-numbered Fedlex laws** (400,405 article rows summed across the parallel DE/FR/IT versions; ~133k articles per language) and a cantonal layer of **15,722 laws** (353,437 articles) covering all 26 cantons via direct portal scraping (LexWork, 18 cantons; SIL, 2 cantons; ZH OpenData, ZH; TI RL, TI), with LexFind PDF fallback supplementing 4 cantons for laws missing from their primary portals. Top-referenced provisions (Table 7, after alias canonicalisation across 20 major DE/FR/IT statute abbreviation pairs) are dominated by federal-court procedural code: *BGG/LTF* Art. 100 (**205,725** decisions), Art. 42 (**201,897**, motivation requirement), and Art. 113 (**122,514**, deadline). Beyond procedural code, *ATSG/LPGA* Art. 61 (social-insurance procedure, 85,207) and *BV/Cst./Cost.* Art. 29 (procedural fundamental rights, 79,115) lead the non-BGG/LTF substantive provisions. The alias canonicalisation is lexical only; temporal validity — mapping a 1998 *OG* reference (the pre-2007 Federal Justice Act, superseded by BGG/LTF) to its current SR location, or distinguishing repealed-vs-current article versions — is a v2.0 deliverable, not addressed in this release.

Materialien graph (Federal Council Botschaften). A distinguishing feature of Swiss civil-law interpretation is the *teleological* reading of statutes via *Materialien* — legislative-history documents (Federal Council messages, parliamentary debates) that record legislative intent. We discover Materialien via the Fedlex JOLUX SPARQL endpoint (`typeDocument = 23`) and ingest them as **5,292 Botschaft documents** (ranging 2003–2026, with DE/FR/IT-parallel coverage), parsed at paragraph granularity (**381,711 paragraphs**) from Akoma Ntoso XML or PDF fallback. We anchor paragraphs to specific (SR, article) pairs by traversing the parliament-chain (`parliamentDraftId` → enacted oc → codified cc → SR), yielding **8,124 article-to-Botschaft links** (DE 2,798, FR 2,742, IT 2,584). The Materialien layer is queryable via `search_materialien` / `get_materialien` MCP tools and surfaces verbatim legislative-intent text under any cited statute article. To our knowledge this is the first open, machine-queryable bridge between the Swiss case-law corpus and the Botschaften that a court would consult in interpretation.

Commentaries. We bridge **362 scholarly commentaries** (CC-BY) from OnlineKommentar.ch and OpenLegalCommentary.ch into the graph, anchored by SR number plus article. Doctrinal layer is currently shallow but growing.

5 Infrastructure

The corpus, graph, statutes, and Materialien are served through four production surfaces:

- **MCP server** (`mcp.opencaselaw.ch/sse`, public SSE). 33 public tools partitioned into four families: retrieval over cases, statutes, commentaries, Materialien, and practice notes; analysis over citations, appeal chains, trends, doctrine, article history, registes, case briefs, and court statistics; generation of exam questions and mock decisions; and verification

Table 5: Statute and doctrinal graph layers. Decision-to-statute edges plus federal/cantonal mirrors plus Materialien (Federal Council Botschaften) and CC-BY commentaries.

Quantity	Value
Decision $\rightarrow$ statute edges	11,261,717
Distinct provisions referenced	283,119
Federal laws (Fedlex SR series)	5,516
Federal articles (DE/FR/IT parallel)	400,405
Cantonal laws (all 26 cantons; 22 direct, 4 LexFind fallback)	15,722
Cantonal articles	353,437
Materialien (Botschaft documents, DE/FR/IT)	5,292
Materialien paragraphs (full-text)	381,711
Article-anchored Materialien links	8,124
Open commentaries (CC-BY)	362

Table 6: Top 15 most-cited decisions in the corpus. Subject area inferred from the decision regeste; in-degree counts citing decisions (with multi-citing pairs counted once per source). Snapshot 2026-05-13.

Decision ID	Reporter form	In-degree
bge_125 V 351	BGE 125 V 351	85,108
bge_122 V 157	BGE 122 V 157	44,233
bge_134 V 231	BGE 134 V 231	36,943
bge_130 V 343	BGE 130 V 343	29,625
bge_126 V 75	BGE 126 V 75	27,703
bge_115 V 133	BGE 115 V 133	26,308
bge_125 V 256	BGE 125 V 256	25,171
bge_125 V 193	BGE 125 V 193	24,057
bge_126 V 353	BGE 126 V 353	23,025
bge_135 V 465	BGE 135 V 465	21,653
bge_137 V 210	BGE 137 V 210	20,421
bge_141 V 281	BGE 141 V 281	19,189
bge_125 V 413	BGE 125 V 413	18,598
bge_130 III 321	BGE 130 III 321	18,421
bge_138 III 374	BGE 138 III 374	18,372

through claim support, attestations, and citation formatting. Two additional maintenance tools are local-only and hidden by the `REMOTE_MODE` filter on the public deployment.

- **REST API** (FastAPI; export endpoints `.docx` / `.pdf` / `.bib` / `.ris` per decision; OpenAPI spec at `/api/openapi.copilot.json` curated for Microsoft Copilot Studio adoption).
- **Word add-in** (`opencaselaw.ch/word`, public). Lawyer-facing client that surfaces the verification rails as inline citations, with structural client+server PII redaction at boundary.
- **Public dashboards** at `https://opencaselaw.ch` (corpus stats, multilingual landing) and `https://opencaselaw.ch/quality.html` (live dataset health).

The retrieval substrate is a SQLite FTS5 index (~63 GB on disk). The nightly refresh uses an *atomic-rebuild* pattern: the new index is constructed in a sibling `.db.tmp` file, then promoted via `os.replace`; live readers (4 uvicorn workers behind nginx) are never interrupted.

Table 7: Top 15 most-referenced statutory provisions. Alias-canonicalised across DE/FR/IT abbreviations of 20 major Swiss federal statutes; decisions citing multiple language forms of the same provision in one document are counted once. The five v1.0 *BGG/LTF* rows now correctly aggregate as a single statute.

Law	Article	Decisions citing
BGG/LTF	Art. 100	205,725
BGG/LTF	Art. 42	201,897
BGG/LTF	Art. 113	122,514
BGG/LTF	Art. 66	97,701
ATSG/LPGA	Art. 61	85,207
BV/Cst./Cost.	Art. 29	79,115
VwVG/PA	Art. 63	70,498
BGG/LTF	Art. 106	65,798
BGG/LTF	Art. 72	64,793
VwVG/PA	Art. 52	64,246
BGG/LTF	Art. 82	62,398
BGG/LTF	Art. 74	62,217
StPO/CP	Art. 428	59,567
BGG/LTF	Art. 95	58,055
ZPO/CPC	Art. 106	56,092

Liveness signal. The deployed MCP endpoint is exercised by external clients (multiple coding-agent and search-engine integrations identified by request user-agent over a four-week window). We summarise aggregate counts in Appendix D and emphasise: adoption is descriptive (the resource is in use), not an evaluation claim about quality. We do not log query payloads.

Traffic pattern across courts. Tier-1 server logs (which include both bot indexers and human visitors) show that *cantonal* jurisprudence receives more decision-page hits than federal jurisprudence: hits to the *Tribunal civil de Genève* (`ge_gerichte`, 252.6k) exceed those to the federal *Bundesgericht* (`bger`, 209.5k); Vaud (`vd_findinfo` + `vd_gerichte`, 100.9k combined), Zürich (`zh_obergericht`, 35.6k), Ticino (`ti_gerichte`, 30.9k), and St. Gallen (`sg_publikationen`, 36.3k) likewise pull substantial traffic. The same rank-ordering persists after a tier-3 bot-vs-browser filter (cantonal > federal in absolute volume), although the bot-stripped totals are lower. The pattern is consistent with a corollary the federation makes empirically testable: cantonal decisions occupy a layer of legal-practice retrieval that a federal-only corpus – the standard scope of prior open Swiss datasets – would miss.

6 Five-rail closing audit

We deploy a five-rail closing audit on every LLM response generated by the public MCP. Each rail addresses a single named error class: (R1) case-citation existence, (R2) statute reference resolution, (R3) verbatim-quote source matching, (R4) decision-date sanity, and (R5) LLM-judge proposition grounding. R1–R4 are deterministic regex+lookup (~3ms total), R5 is judge-mediated (~2s, ~\$0.005). The decomposition by error class lets each rail be tested, ablated, and improved in isolation.

The closing audit operates on a draft answer D and a retrieval-state pool S (decisions and statutes the generator had access to). It returns a list of issues $\mathcal{I}(D, S)$, each labelled with one of five named error classes. The five rails operate *in parallel* on the same draft.

Rail 1 — case-citation existence. A regex parses the draft for Swiss case-citation forms (BGE / ATF / DTF, BGer / TF dockets, BVGer / BStGer / BPatGer dockets, MKGE numer-

ical references). Each parsed citation is resolved against the corpus by exact-match lookup with a strict canonical-prefix guard. Latency 2–5 ms regardless of citation count. Catches: invented BGE numbers, mis-typed docket numbers, citations to retracted decisions. An optional pin-point check requires `E.n.m` or `consid.n.m` markers to match the structured-extraction sidecar (`decision_structure.db`) or the body `regeste` of the cited decision.

Rail 2 — statute-reference resolution. A second regex parses for `Art.NLAW` forms (with German / French / Italian abbreviation synonyms; LTF=BGG, CO=OR, CC=ZGB, etc.) and resolves each against the Fedlex SPARQL mirror cached locally as `statutes.db`. Two-step resolution: abbreviation → SR number → article text, cached at process scope. Latency 1–3 ms. Catches: fabricated law abbreviations, wrong article numbers, references to repealed articles. Connector words (“und”, “oder”, “et”, “ou”) are case-sensitively filtered to prevent the regex eating them as fake law codes.

Rail 3 — verbatim-quote source-matching. Quoted substrings of ≥ 30 characters in the draft are extracted (German low-9 “”, French «...», Italian «...», English curly and ASCII variants) and required to appear verbatim — after whitespace + quote-mark normalisation — in the source pool *S*. The pool comprises the `regeste` text plus the first 8 k characters of each cited decision, plus the article text of each cited statute. Latency ~ 2 ms. Catches: hallucinated quotations, paraphrases inside quotation marks.

Rail 4 — decision-date sanity. For each verified case citation, parse adjacent `vom / du / del DD.MM.YYYY` constructs within 60 characters of the citation span and compare to the stored `decision_date`. Latency < 1 ms. Catches: model emits a real BGE but pairs it with a wrong date, suggesting confusion between two cases with similar dockets.

Rail 5 — proposition grounding (opt-in, judge-mediated). For each verified citation, a claim sentence is extracted (preceding sentence, with forward-fallback for “In BGE X, ...” patterns). “See”-cite markers (`vgl.`, `cf.`, `siehe`, `cfr.`) suppress the rail. All (claim, source) pairs go to a separate LLM judge call (same model family as the generator) that returns one of {`yes`, `partial`, `no`, `contradicts`, `unrelated`}. Verdicts in {`no`, `contradicts`, `unrelated`} become issues. One judge call per draft regardless of citation count. Latency 1.5–2.5 s. Cost $\approx$ \$0.005 per call.

Output canonicalisation. When all rails pass, the audit emits a `linked_text` field with every parsed citation replaced by its corpus-canonical form wrapped in a Markdown link to `mcp.opencaselaw.ch/entscheid/<id>`. Drafts arrive with messy citation spacing or case and leave publication-ready.

Empirical baseline (from prior-only condition). On the v2 30-question Swiss Legal RAG bench, ~ 33 distinct authority citations were emitted across 30 drafts; the deterministic rails (R1–R4) flagged 1 (3 % observed fabrication rate). For context, Magesh et al. [7] report 17–33 % on commercial legal-RAG and Dahl et al. [3] report 58–88 % on general-purpose LLMs (ChatGPT-4 to Llama-2) in different settings; we cite these to bracket the literature, not as a like-for-like comparison. The full five-rail audit including the LLM-judge rail R5 reaches a 66.7 % wrong-draft flag rate at a 41.7 % false-positive rate; calibration of the judge’s threshold remains an open problem and the rail should be deployed in advisory mode pending future calibration work.

Retrieval-augmented condition. A retrieval-augmented re-run on the same 30-question bench is the highest-priority follow-on for v1.1 of the benchmark; we publish it as a versioned update on the bench’s Hugging Face page when ready.

Table 8: Search-quality baseline by query language. 100-question golden set, offline (BM25+RRF, no LLM rerank), 2026-03-19. Italian queries score highest because the Italian corpus is the smallest (lowest target ambiguity). The “Unknown” row covers 3 queries whose query language was not classified at evaluation time and are reported separately for completeness.

Language	n	MRR@10	Recall@10	Hit@1	nDCG@10
DE	74	0.431	0.422	0.284	0.463
FR	16	0.510	0.630	0.375	0.611
IT	7	0.770	0.893	0.714	0.803
Unknown	3	0.500	0.667	0.333	0.945
Total	100				

Table 9: Cross-lingual leading-case retrieval. Cells give MRR@10 (n queries) for retrieval of a leading-case decision when the query is in language Q and the decision’s authoritative language is T . 50 leading cases (in-degree range 4,412–53,251, median 8,180; top 0.025 % of cited decisions), with regeste-derived parallel queries in DE/FR/IT. BM25+RRF (no LLM rerank). The 50 cases include 38 originally DE and 12 originally FR; none is originally IT (consistent with the corpus distribution among most-cited decisions). Overall MRR@10=0.630, Hit@10=0.833 across n=150 queries. Numbers are upper-bounded by query–regeste vocabulary coupling; see §11.

Q lang ↓ T lang →	DE	FR	IT
DE	0.617 (n=38)	0.333 (n=12)	—
FR	0.566 (n=38)	0.785 (n=12)	—
IT	0.776 (n=38)	0.549 (n=12)	—

7 Cross-lingual retrieval diagnostic (regeste-derived)

Setup. We select 50 leading cases from the most-cited subset of the corpus (in-degree range 4,412–53,251, median 8,180, measured at benchmark-construction time; the citation graph has since grown — see Table 2 and Table 6 for current values), stratified to at most four cases per legal area to ensure diversity across the 22 covered areas (constitutional, social-insurance, federal-court procedure, criminal, civil-procedure, contract, tax, asylum, and more). All 50 cases lie in the top 0.025 % of distinct cited decisions in the corpus by in-degree. For each case we extract one query per language (DE, FR, IT) from terms in that language’s official regeste, after dropping article-reference noise (e.g. *Art. 6 Abs. 1 UVG*) and language-specific stop words. The query construction is keyword-list style, not a free natural-language paraphrase; we discuss the realism implications in Section 11. The dataset is released as `benchmarks/swiss_legal_rag_bench/cross_lingual_v1.jsonl` (CC0). Of the 50 cases, 38 are originally German and 12 originally French — consistent with the corpus distribution among most-cited decisions. No selected case is originally Italian; the absent Italian-target column is therefore a sampling limitation, not a failed retrieval cell, and we treat it as a coverage gap (Section 11). *The benchmark is single-curator; multi-annotator agreement is deferred to v2.0, and we read the absolute scores as an upper bound on realistic retrieval performance (Section 11).*

Results. Table 9 reports MRR@10 and Table 10 reports Hit@10 per (Q lang, T lang) cell. Overall MRR@10=0.630, Hit@10=0.833 (125 of 150 queries place the target in the top 10).

Findings.

- **Italian queries retrieve German targets best (MRR=0.776).** Italian queries outperform German queries on the *same* 38 German targets (DE→DE MRR=0.617). One

Table 10: Cross-lingual leading-case retrieval, Hit@10. Same setup as Table 9. Hit@10 = fraction of queries where the target leading case appears in the top 10 results.

Q lang ↓ T lang →	DE	FR	IT
DE	0.789	0.333	—
FR	0.763	1.000	—
IT	1.000	1.000	—

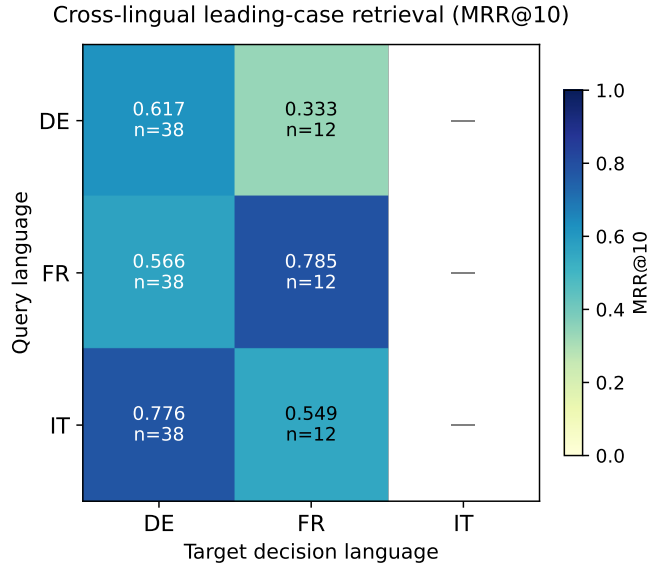


Figure 1: Cross-lingual leading-case retrieval (MRR@10) heatmap. Rows are query languages, columns are target authoritative-version languages.

mechanism: Italian regestes carry distinctive shared markers (*DTF*, *cpv.*, *LAINF*) that anchor retrieval, while German queries compete with the broader German subset (449k decisions, the largest stratum) and so retrieve more in-language distractors.

- **Same-language French is the strongest cell (MRR=0.785, Hit@10=1.0).** For the 12 French-original leading cases, French queries place the target in the top 10 every time and at rank 1 in 67 % of cases.
- **German-to-French is the hardest cell (MRR=0.333, Hit@10=0.333).** German queries struggle to retrieve French-original BGEs because the discriminative German terms in the regeste-derived queries do not appear in the French-language full text and the BM25+RRF baseline does no cross-lingual translation.
- **Diagonal vs. off-diagonal gap is small.** Weighted across n, same-language diagonal cells (DE→DE n=38, FR→FR n=12) score MRR=0.657; the four off-diagonal cells (n=100 total) score MRR=0.616. The 0.04 gap shows that cross-lingual retrieval against a multilingual leading-case index is feasible without translation, although the cell-by-cell variance (0.333 to 0.776) is large.

8 Bench: prior-only vs. retrieval-augmented

We evaluate end-to-end draft quality on a 30-question Swiss Legal RAG bench (`benchmarks/swiss_legal_rag_bench/questions.jsonl`; spans 17 substantive areas; languages DE=18,

Table 11: Swiss Legal RAG bench results, both conditions, side-by-side. 30 questions (DE=18, FR=8, IT=4). Generator and judges: Claude Sonnet 4.6. Retriever: live MCP `/api/decisions`, top-K=5 with annotated statute text injected. Outcome classes follow Butler & Butler 2026: *correct* (correctness $\wedge$ groundedness $\wedge$ retrieval); *hallucination* ($\neg$ groundedness); *retrieval* (grounded $\wedge$ $\neg$ retrieval $\wedge$ $\neg$ correctness); *reasoning* (grounded $\wedge$ retrieval $\wedge$ $\neg$ correctness). In prior-only mode the retriever is skipped, so only *correct* / *hallucination* apply.

Metric	Prior-only	RAG-aug
n questions	30	30
Correctness (entailment judge)	86.7%	63.3%
Groundedness (passage-support judge)	—	93.3%
Retrieval accuracy (annotated case in top-5)	—	53.3%
<i>Error-class decomposition (n decisions per class)</i>		
correct	26	17
hallucination	4	2
retrieval	—	9
reasoning	—	2

FR=8, IT=4; each question paired with a corpus-anchored reference answer plus annotated statute and where applicable a leading-case `decision_id`. We adopt the Butler & Butler 2026 [2] correctness $\times$ groundedness $\times$ retrieval-accuracy decomposition (4 outcome classes: *correct* / *hallucination* / *retrieval* / *reasoning*). Generator and judges are Claude Sonnet 4.6; retriever is the live MCP `/api/decisions` endpoint with top-K=5 and annotated statute text injected. Reproducible via `benchmarks/swiss_legal_rag_bench/evaluate.py`.

Conditions and results. Table 11 reports both conditions side-by-side. The *prior-only* condition skips retrieval entirely: the generator answers from training prior, with no passages provided. The *RAG-augmented* condition runs the full pipeline (retrieve top-5 cases + inject annotated statute text + answer constrained to retrieved passages).

Counter-intuitive finding. The prior-only condition scores higher on correctness (86.7%, 26/30) than the retrieval-augmented condition (63.3%, 19/30). The gap is driven not by hallucinations — groundedness in the RAG-augmented condition is 93.3% (28/30) — but by retrieval failures: only 53.3% (16/30) of questions had the annotated leading case in the top-5 retrieved set, and the strict “answer using ONLY the retrieved passages” prompt blocks the generator from falling back to its (substantial) prior knowledge of Swiss federal law. Of the 13 RAG-aug failures, 9 are *retrieval* class (the model honestly used what was retrieved but the right case wasn’t in it), 2 are *reasoning* (right context, wrong conclusion), and only 2 are true *hallucination* (claimed something not in the passages).

Interpretation. For a model with Sonnet 4.6’s prior coverage of Swiss federal law, naive RAG-augmentation is a net regression on this bench unless one of (a) retrieval precision exceeds the generator’s prior accuracy, or (b) the prompt allows fallback to prior knowledge when retrieval comes back empty-handed. We do *not* claim a general hallucination-rate result. The 13.3% prior-only hallucination rate (4/30, all in the DE stratum) is below commercial-legal-RAG numbers reported by Magesh et al. [7] (17–33%) and far below the general-LLM range from Dahl et al. [3] (58–88%), but the conditions, jurisdictions, models, and grading criteria all differ; we cite to bracket the literature, not as a like-for-like comparison.

The five-rail audit (Section 6) is condition-agnostic — the deterministic rails R1–R4 make no assumption about whether a draft was produced with or without retrieval, so the same

deployment audits both pipelines. Rail-specific empirical numbers (e.g. R1 unresolvable-citation rate, R5 judge false-positive rate) are reported with the audit-pipeline description in Section 6; we do not duplicate them here.

Retrieval-augmented condition. A side-by-side retrieval-augmented re-run is the highest-priority follow-on for v1.1. We expect retrieval to substantially reduce R1 fabrication-existence flags (the system retrieves real authorities rather than relying on prior); the open question is whether judge-rail false-positive rate increases (more authorities to ground against) or decreases (clearer source pool). We commit to publish the retrieval-augmented numbers as a versioned update on the benchmark’s Hugging Face page rather than holding the resource paper’s release on it.

9 Dataset health and drift

OpenCaseLaw operates a four-layer quality framework:

L1 Function tests (575 pytest cases across 58 modules, every CI push, ~ 90 s).

L2 Pipeline guards (5 codified `_normalize_*` passes in `build_fts5.py`, every nightly rebuild).

L3 Dataset audit (63 `check_*` functions across 17 axes: schema, dates, dockets, courts, languages, regeste coverage, URL integrity, duplicates, text quality, structure, cross-DB consistency, citation graph, statute graph, floors, per-court drift, exports round-trip, MCP-tool round-trip).

L4 Production smoke (every 5 min via systemd timer; 6 endpoint probes; $p_{95} < 2$ s).

A dataset-audit run that detects a critical regression *blocks the nightly publish*: the orchestrator (`publish.py`) refuses to advance to the git-push step and instead emits an urgent ntfy alert. Critical conditions include a total-count cliff $> 1\%$, schema break, EGMR re-regression, court not in registry, future date count > 50 , cross-DB row-count mismatch $> 0.5\%$. Warning conditions (per-court drift $> 10\%$ from 7-day median, OCR-quality regression, citation-graph edge drop $> 5\%$) emit alerts but do not block publish.

Operationally, production changes are governed by a committed decision-rules document (`docs/decision_rules.md`) that requires runbooks for production changes, manual operator review before any public rollback, Saturday-only schema changes, and measurable dashboard backing for public claims.

The framework codifies real incidents. Three case studies illustrate the pattern of *audit detects* $\rightarrow$ *fix ships* $\rightarrow$ *check codifies*:

Case study 1 — the SG merge anomaly (alphabetical-collision class). On 2026-04-29 a routine total-count check fired: the `sg_publikationen` court had lost 1,450 rows ($\approx 90\%$ loss) on the nightly rebuild without any scraper failure. The diagnostic chain was: (a) the `INSERT OR IGNORE` dedup pattern in `build_fts5.py` processes JSONL shards in alphabetical order; (b) the `entscheidsuche` side-feed (`es_sg_publikationen.jsonl`) processes *before* the direct shard (`sg_publikationen.jsonl`) because ‘e’ precedes ‘s’; (c) the two shards happen to construct identical decision-id prefixes (`sg_publikationen_<docket>`); (d) the earlier-processed shard wins all dedups, silently dropping the chamber-correct rows from the direct shard. Reproducer scripts (`scripts/sg_anomaly_*.py`) confirmed the hypothesis on a fresh database. Two fixes shipped on 2026-04-30: (i) the problematic `es_sg_publikationen` feed was retired (commit `f249f1f`); (ii) an architectural fix forced direct shards to process before `es_*` shards (commit `713afe3`, gated by `BUILD_FTS5_DIRECT_FIRST=1`). The pattern check across all multi-chamber

cantons confirmed only SG was affected. The fix added two permanent regression checks to the L3 audit: per-court row-count drift >10 % (warning) and per-court row-count drop >50 % (critical, blocks publish).

Case study 2 — the EGMR re-regression class (cross-source double-coverage). The `entscheidsuche CH_BGE` feed includes `cedh` (ECHR) decisions that should land under our `bge_egmr` court rather than `bge`. Without an override, 474 ECHR cases were duplicated with divergent metadata (BGE side carried the full text but no chamber labels, the BGE-EGMR side carried chamber and regeste but truncated text). External audit on 2026-04-29 surfaced this; we backfilled production with a best-of-both merge (preserve full text from BGE side, preserve chamber and regeste from BGE-EGMR side) and deleted the BGE side. The L3 audit now includes a hard-zero check on (`court='bge'` AND `source_url` contains `cedh`): *any* positive count blocks the next publish.

Case study 3 — the König P-class incidents. A second external audit (König, 2026-04-29 to 2026-05-01) uncovered seven distinct issue classes over two days: oversized HUDOC regestes (P1, full judgment text in the regeste field, max 165k chars; truncation pass added in `build_fts5._truncate_oversized_regestes`); short-text fragments that belonged in the regeste field (P7, migration pass added); internal-newline dockets (5,441 rows fixed via `extended_normalize_dockets`); BS/GL relative source URLs (694 rows, host prefix recovery via `_normalize_source_urls`); and ZH targeted gap-fill of 11 missing dockets. Each issue class is now a permanent `check_*` function in `quality/checks/*.py` and a build-time auto-correction pass in `build_fts5.py`.

Detection cadence. During the codification period (2026-04-29 to 2026-05-01), the L3 audit fired at least one critical condition on the majority of nightly runs because each external-audit finding initially manifested as a publish-blocking regression. After each issue class was codified into a build-time auto-correction pass in `build_fts5.py`, the same regression class was prevented at build time on subsequent runs, so the audit no longer fires for it. The drift band ($\pm 5 \times \text{MAD}$ on a 7-day rolling median) flags court-level anomalies typically within one nightly run of the underlying scrape change; a deeper longitudinal study of detection latency is left for future work.

10 Discussion and generalisation

The cross-lingual results have a clear takeaway for legal-RAG practitioners: *retrieval quality is asymmetric across languages and direction*. A naive deployment that only validates same-language MRR will overestimate user-facing quality in mixed-language jurisdictions. Our public benchmark allows quantitative pre-deployment audits of cross-lingual performance.

The verification framework generalises beyond Switzerland. Each rail of the closing audit is corpus-agnostic given (a) an enumerable decision corpus, (b) a statute mirror, and (c) an extraction grammar for the local citation conventions. We sketch an English-law application in Appendix B. The *Materialien* layer is the most directly Switzerland-specific contribution: jurisdictions with comparable legislative-history doctrine (Germany’s *Gesetzgebungsmaterialien*, Canada’s *Hansard*, the United States’ *legislative history*) could replicate the parliament-chain SPARQL traversal pattern over their own gazette ontologies.

The early-adoption signal — a Swiss-French law firm and a cantonal-government legal department actively querying the deployed MCP within weeks of soft launch — suggests that the resource fills a deployment-blocker gap rather than only a research-evaluation gap. The corollary risk: any infrastructure regression now affects production legal research, which motivates the four-layer QC framework (Section 9) as a non-negotiable accompaniment to release.

11 Limitations and ethics

Benchmark scope. The cross-lingual question set is single-curator and limited to 50 cases (150 trials). It is anchored on the citation graph (target decisions are objectively-ranked leading cases) but the queries themselves are programmatic keyword extractions from the corresponding language’s regeste, not free legal-research paraphrases. v2.0 of the benchmark will recruit multi-annotator panels to reach inter-annotator agreement.

Query–target coupling (upper-bound caveat). Because each query is derived from the same case’s official multilingual regeste, the query and target share distinctive vocabulary the retrieval system can latch onto. A real legal-research query, written without sight of the regeste, may not contain those discriminators. We therefore read the absolute MRR/Hit@10 numbers as an *upper bound* on realistic retrieval performance, and the cross-cell ranking (which cells are easier or harder, by how much) as the more reliable signal. v2.0 will replace regeste-derived queries with held-out, lawyer-authored research questions.

Italian-target coverage. No top-50 case is originally Italian (top-50 is dominated by social-insurance and procedural BGEs that originate in the German-language stream). This is consistent with the corpus distribution among most-cited decisions but is a coverage gap; v2.0 will explicitly include Italian-original cases sampled at lower citation centrality.

Privacy. Court decisions in Switzerland are public acts. Pseudonymisation practice varies across cantons and even across the same court’s chambers; the corpus inherits whatever pseudonymisation the source decision carries. We do not de-pseudonymise, and we also do not re-pseudonymise material that the issuing court already published un-pseudonymised. We support take-down requests through the contact channel listed on the public dashboard.

Calibration. The grounding-judge rail (R5) is a judge-mediated LLM call; we report a 41.7% false-positive rate on correct drafts (the figure cited consistently in Section 6 and `sections/audit_pipeline.tex`). Full-binary deployment would over-flag; we recommend the rail be used in advisory mode pending v2.0 calibration work.

12 Open access and artifacts

All artifacts are publicly available:

- Corpus: Hugging Face <https://huggingface.co/datasets/voilaj/swiss-caselaw> (CC0)
- Benchmark: Hugging Face <https://huggingface.co/datasets/voilaj/swiss-legal-rag-bench> (CC0)
- Code: <https://github.com/jonashertner/caselaw-repo-1> (MIT)
- Live MCP: <https://mcp.opencaselaw.ch> (SSE; OpenAPI at `/api/openapi.copilot.json`)
- Word add-in (lawyer-facing client): <https://opencaselaw.ch/word>
- Dashboards: <https://opencaselaw.ch>, <https://opencaselaw.ch/quality.html>

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A Reproducibility manifest

All claims in this paper are backed by code, data, and configuration that anyone can clone or query. This appendix lists every claim’s provenance so a reviewer can verify in less than five minutes.

Code.

- **Repository:** `https://github.com/jonashertner/caselaw-repo-1` (MIT licence). The release tag for this paper is `paper-resource-2026-05` (to be assigned at submission; prior to that, all numerical claims trace to the commit pinned in the HuggingFace dataset’s `manifest.json`).
- **Five-rail audit pipeline:** `mcp_server.py`, function `_handle_attest_response`. The four deterministic rail implementations are inline in the same file: `_audit_statutes` (R2), `_audit_quotes` (R3), `_audit_dates` (R4), `_audit_grounding` (R5). R1 (case-citation existence) is inline in `_handle_attest_response`.
- **Quality framework:** `quality/` package (`runner.py`, `cli.py`, `drift.py`, `baseline.py`, `smoke.py`, plus 17 modules in `quality/checks/` containing 63 `check_*` functions).
- **Cross-lingual harness:** `benchmarks/run_cross_lingual.py`.
- **Pipeline guards:** `build_fts5.py` `_normalize_*` / `_dedup_*` / `_truncate_*` passes.

Data artifacts.

- **Corpus:** HuggingFace `voilaj/swiss-caselaw` (CC0). Each release is accompanied by a `manifest.json` listing per-court row counts plus SHA-256.
- **Cross-lingual benchmark v1:** HuggingFace `voilaj/swiss-legal-rag-bench/cross_lingual_v1.jsonl` (CC0). 150 queries, anchored on the citation graph.
- **Audit-bench v2:** same repository, `verification_seed_v2.jsonl`.
- **Citation graph:** derived from the corpus via `search_stack/build_reference_graph.py`; release snapshot available on Zenodo (DOI to be assigned at submission).
- **Statute mirrors:** `statutes.db` (federal, derived from Fedlex SPARQL endpoint), `cantonal_laws.db` (cantonal, derived from LexWork+SIL+ZH portals via `scrapers/cantonal_laws/`).

Live system (runtime verification).

- <https://mcp.opencaselaw.ch/sse> (MCP server endpoint).
- <https://opencaselaw.ch/> (corpus dashboard, multilingual).
- <https://opencaselaw.ch/quality.html> (live dataset health).
- <https://opencaselaw.ch/coverage/> (per-source scrape health).

Reproducing key numerical claims. Two modes:

Offline (recommended for archival reproducibility):

```
git clone https://github.com/jonashertner/caselaw-repo-1
cd caselaw-repo-1
make verify-offline
```

Reads only from JSON snapshots committed at the paper-resource tag: the corpus-size, citation-graph resolution rate, and cross-lingual MRR/Hit@10 all come from files under `docs/paper/v3/tables/` and `benchmarks/swiss_legal_rag_bench/results/`. No network access required. Runs in ~5s. Expected output: “All headline claims reproduce against committed snapshots”.

Live (additionally verifies the deployed system still serves the same data):

```
make verify
```

Same as `verify-offline` except corpus size is read from `/health` on the live MCP endpoint. Floors a corpus-shrink regression.

Containerised reviewer image:

```
make docker-build
docker run --rm opencaselaw-reviewer:<date> make verify-offline
```

Builds a ~200 MB Python 3.12 + SQLite image (see `Dockerfile.reviewer`) that contains only the committed snapshots and audit scripts. Confirms the headline numbers reproduce in a clean sandbox without depending on the reviewer’s local Python toolchain.

For deeper verification (full citation-graph rebuild and benchmark re-runs against the live corpus):

```
git checkout paper-resource-2026-05
pip install -r requirements.txt
huggingface-cli download voilaj/swiss-caselaw --local-dir output/
python -m quality.cli run # 4-layer dataset health
python benchmarks/run_search_benchmark.py # Tab.~2 baseline
python benchmarks/run_cross_lingual.py # Tab.~3, Fig.~2 (cross-lang)
python -m benchmarks.citation_resolution_analysis
make test # full pytest suite (~40 s)
```

Telemetry, costs, latency. The MCP server exposes `/health` (corpus row count) and is monitored every 5 minutes by a systemd timer that probes 6 endpoints (search, decision, export docx/pdf/bib/ris) with $p_{95} < 2$ s. Every audit run since 2026-05-01 is logged with full per-rail latency and judge cost to `logs/audit_telemetry.jsonl`.

Hardware. Production runs on a Hetzner CCX43 dedicated instance (16 vCPUs, 64 GB RAM, 200 GB attached volume). The full nightly publish takes 3 h 50 min, dominated by the FTS5 rebuild.

Licences.

- Code: MIT.
- Corpus packaging + benchmark: CC0 to the extent rights exist. Court decisions in Switzerland are official published acts.
- Commentaries: CC-BY-4.0 (OnlineKommentar.ch) / CC-BY-SA-4.0 (OpenLegalCommentary.ch).
- Federal statutes: derived from Fedlex SPARQL endpoint; Swiss Confederation copyright applies, fair-use distribution permitted.

B English-law generalisation sketch

Goal. Demonstrate that the five-rail closing-audit decomposition is corpus-agnostic, given (a) an enumerable decision corpus, (b) a statute mirror, and (c) a citation extraction grammar for the local conventions. We sketch this for English law (UK + Welsh decisions, BAILII).

Components. Citation grammar. English neutral citations follow a fixed schema: [YEAR] COURT NN, where COURT is one of a small controlled vocabulary (UKSC, EWCA Civ, EWCA Crim, EWHC, UKHL, UKPC, EWFC, EWCOP, etc.). A regular expression captures these reliably:

```
r"\\[(\\d{4})\\]\\s+(UKSC|UKHL|UKPC|EWCA\\s+Civ|EWCA\\s+Crim  
|EWHC|EWCOP|EWFC|EWCC|UKEAT|UKUT|UKFTT)\\s+(\\d+)"
```

Statute mirror. `legislation.gov.uk` publishes machine-readable XML for every UK Public General Act since 1267. The R1 statute-existence rail can be implemented identically to our Swiss Fedlex mirror, swapping the `statutes.db` schema in place.

Decision corpus. BAILII publishes UK and Irish decisions free of charge (single-source: subscription model retired, donate-as-you-go). The corpus is approximately 750,000 decisions across UK courts. A single-script crawler is feasible.

What works without modification.

- **R1 case-citation existence:** one regex change (above).
- **R3 verbatim-quote source matching:** identical algorithm, re-uses our 5-gram shingle hashing pipeline.
- **R4 decision-date sanity:** identical ($\text{date} \leq \text{today}$ and $\geq \text{corpus floor}$).
- **R5 grounding judge:** identical (LLM-mediated, prompt templates have only the corpus name in them).

What requires adaptation.

- **R2 statute resolution:** needs the `legislation.gov.uk` URN scheme rather than Fedlex SR numbers. URN structure is well-documented; estimated 1–2 weeks of integration work.
- **Cross-lingual:** not applicable (English law is monolingual). The cross-lingual benchmark would target a different multilingual jurisdiction (Belgium, Canada-Quebec, South Africa, India).

Status. The architecture above is a sketch only; we have not yet performed an English-law evaluation. Full implementation and evaluation are left for future work. The two implementation tasks estimated above (citation-grammar swap, statute-mirror swap) are the non-trivial portions; the corpus crawl and rail orchestration reuse this paper’s harness directly.

C Cross-lingual question set

Schema. Each row is a leading case (anchored on the citation graph) with a parallel query authored in each of DE/FR/IT. The full release at `benchmarks/swiss_legal_rag_bench/cross_lingual_v1.jsonl` also carries: `q_id`, `target_lang`, `anchor_terms`, `difficulty`.

Table 12: Cross-lingual question set (page 1 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 125 V 351	accident_insurance	DE	Unfallversicherungsgesetz Verbindung VwVG Beweiswürdigung Parteigutachten
BGE 125 V 351	accident_insurance	FR	LAA liaison appréciation preuves expertise
BGE 125 V 351	accident_insurance	IT	LAINF Cost relazione valutazione prove
BGE 134 V 231	accident_insurance	DE	Unfallversicherungsgesetz Beweiswert diagnostischer Methoden funktionelle
BGE 134 V 231	accident_insurance	FR	LAA valeur probante méthodes diagnostiques
BGE 134 V 231	accident_insurance	IT	LAINF valore probatorio metodi diagnostici
BGE 122 V 157	constitution	DE	Bundesverfassung umfassen formellen Anspruch Beizug
BGE 122 V 157	constitution	FR	Cst. permettent déduire droit caractère
BGE 122 V 157	constitution	IT	Cost. Cost essere dedotto diritto
BGE 130 V 343	general_insurance_law	DE	ATSG seit Januar geltenden Fassung
BGE 130 V 343	general_insurance_law	FR	LPGA teneur vigueur janvier relation
BGE 130 V 343	general_insurance_law	IT	LPGA versione vigore gennaio relazione
BGE 138 III 374	civil_procedure	DE	Zivilprozessordnung Beweiserhebung Berufungsverfahren Begründung Berufung
BGE 138 III 374	civil_procedure	FR	CPC administration preuves appel motivation
BGE 138 III 374	civil_procedure	IT	CPC assunzione prove appello motivazione
BGE 133 II 249	federal_court_procedure	DE	Bundesgerichtsgesetz öffentlich-rechtlichen Angelegenheiten Baubewilligung Nachbarbeschwerde
BGE 133 II 249	federal_court_procedure	FR	LTF relation matière droit public
BGE 133 II 249	federal_court_procedure	IT	LTF relazione materia diritto pubblico
BGE 125 V 256	disability_insurance	DE	Invalidenversicherungsgesetz Verbindung Massgebender Hilfsigkeitsgrad Veränderungen
BGE 125 V 256	disability_insurance	FR	LAI relation Degré impotence déterminant
BGE 125 V 256	disability_insurance	IT	LAI relazione Grado grande invalidità
BGE 141 V 281	general_insurance_law	DE	ATSG Verbindung insbesondere psychosomatische Leiden
BGE 141 V 281	general_insurance_law	FR	LPGA corrélation particulier affections psychosomatiques
BGE 141 V 281	general_insurance_law	IT	LPGA relazione particolare affezioni psicosomatiche
BGE 126 V 75	disability_insurance	DE	Invalidenversicherungsgesetz Kürzung Tabellenlöhnen Bestimmung Invalideneinkommens
BGE 126 V 75	disability_insurance	FR	LAI Réduction salaires ressortant statistiques
BGE 126 V 75	disability_insurance	IT	LAI Riduzione salari statistici fini
BGE 133 V 108	disability_insurance	DE	Invalidenversicherungsgesetz Kraft gewesen Dezember Massgebende
BGE 133 V 108	disability_insurance	FR	LAI vigueur jusqu décembre Base
BGE 133 V 108	disability_insurance	IT	LAI vigore fino dicembre Criterio

Table 13: Cross-lingual question set (page 2 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 126 V 353	accident_insurance	DE	Unfallversicherungsgesetz Arbeitswegunfall Teilzeitbeschäftigten Frage Arbeitspensum
BGE 126 V 353	accident_insurance	FR	LAA Accident trajet victime personne
BGE 126 V 353	accident_insurance	IT	LAINF Infortunio tragitto vittima persona
BGE 115 V 133	accident_insurance	DE	Unfallversicherungsgesetz Präzisierung veröffentlichten adäquaten Kausalzusammenhang
BGE 115 V 133	accident_insurance	FR	LAA Précision apportée publiée relative
BGE 115 V 133	accident_insurance	IT	LAINF Precisazione concernente nesso causalità
BGE 125 V 413	disability_insurance	DE	Invalidenversicherungsgesetz VwVG Verbindung Streitgegenstand Begriff
BGE 125 V 413	disability_insurance	FR	LAI relation objet litige Notion
BGE 125 V 413	disability_insurance	IT	LAI relazione oggetto lite Nozione
BGE 125 V 193	unemployment_insurance	DE	Arbeitslosenversicherungsgesetz Einstellung Anspruchsberechtigung Verlust Anspruchs
BGE 125 V 193	unemployment_insurance	FR	LACI suspension déchéance droit indemnité
BGE 125 V 193	unemployment_insurance	IT	LADI sospensione decadenza diritto indennità
BGE 136 I 229	federal_court_procedure	DE	Bundesgerichtsgesetz Anfechtung Prüfungsergebnisses hier Masterabschluss
BGE 136 I 229	federal_court_procedure	FR	LTF contre résultat examen espèce
BGE 136 I 229	federal_court_procedure	IT	LTF Cost contro esito esame
BGE 130 III 321	civil_code	DE	Zivilgesetzbuch Eintritt Versicherungsfalls Beweis Beweislast
BGE 130 III 321	civil_code	FR	CC survenance sinistre preuve Fardeau
BGE 130 III 321	civil_code	IT	CC sopravvenienza sinistro prova Onere
BGE 132 V 93	general_insurance_law	DE	ATSG VwVG Begutachtung durch Sozialversicherer
BGE 132 V 93	general_insurance_law	FR	LPGA Expertise ordonnée assureur social
BGE 132 V 93	general_insurance_law	IT	LPGA Perizia ordinata dall assicuratore
BGE 134 II 244	federal_court_procedure	DE	Bundesgerichtsgesetz Nichteintreten ungenügend begründete setzt
BGE 134 II 244	federal_court_procedure	FR	LTF Irrecevabilité insuffisamment motivé exige
BGE 134 II 244	federal_court_procedure	IT	LTF Inammissibilità motivato modo sufficiente
BGE 134 I 140	echr	DE	EMRK öffentlich-rechtlichen Angelegenheiten Schutzmassnahmen gegen
BGE 134 I 140	echr	FR	CEDH zurichoise protection contre violence
BGE 134 I 140	echr	IT	CEDU segg Cost legge zurighese
BGE 127 I 38	criminal_procedure	DE	Strafprozessordnung UNO-Pakt Unschuldsvermutung Beschränkung Kognition
BGE 127 I 38	criminal_procedure	FR	CPP aCst Pacte présomption innocence
BGE 127 I 38	criminal_procedure	IT	CPP Cost vCost Patto presunzione

Table 14: Cross-lingual question set (page 3 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 140 III 86	federal_court_procedure	DE	Bundesgerichtsgesetz Pflicht Begründung Rechtsverletzungen Willenserklärung
BGE 140 III 86	federal_court_procedure	FR	LTF obligation motiver violations droit
BGE 140 III 86	federal_court_procedure	IT	LTF obbligo motivare violazioni diritto
BGE 130 V 445	general_insurance_law	DE	ATSG Intertemporalrechtliche Anwendbarkeit materieller Bestimmungen
BGE 130 V 445	general_insurance_law	FR	LPGA Application intertemporelle dispositions matérielles
BGE 130 V 445	general_insurance_law	IT	LPGA Applicabilità intertemporale disposizioni materiali
BGE 137 I 195	constitution	DE	Bundesverfassung Anspruch rechtliches Gehör Replikrecht
BGE 137 I 195	constitution	FR	Cst. droit être entendu réplique
BGE 137 I 195	constitution	IT	Cost. Cost diritto essere sentito
BGE 140 III 115	international_private_law	DE	IPR-Gesetz zweiter Spiegelstrich LugÜ Internationale
BGE 140 III 115	international_private_law	FR	LDIP second tiret Compétence internationale
BGE 140 III 115	international_private_law	IT	LDIP secondo trattino CLug Competenza
BGE 129 I 8	constitution	DE	Bundesverfassung Genfer Reglements April über
BGE 129 I 8	constitution	FR	Cst. Règlement genevois avril fixant
BGE 129 I 8	constitution	IT	Cost. Cost Regolamento ginevrino aprile
BGE 143 IV 241	criminal_procedure	DE	Strafprozessordnung StPO häusliche Gewalt Einstellung
BGE 143 IV 241	criminal_procedure	FR	CPP violence domestique classement pénale
BGE 143 IV 241	criminal_procedure	IT	CPP violenza domestica abbandono procedimento
BGE 141 IV 61	criminal_code	DE	Strafgesetzbuch StGB Mord Konkurrenz mehreren
BGE 141 IV 61	criminal_code	FR	CP assassinat concours entre assassinats
BGE 141 IV 61	criminal_code	IT	CP assassinio concorso reati Riepilogo
BGE 142 II 218	constitution	DE	Bundesverfassung StAhiG VwVG Amtshilfe Steuersachen
BGE 142 II 218	constitution	FR	Cst. assistance administrative matière fiscale
BGE 142 II 218	constitution	IT	Cost. Cost assistenza amministrativa materia
BGE 126 V 319	health_insurance	DE	Krankenversicherungsgesetz Behandlung unfallbedingter Schädigungen Kausystems
BGE 126 V 319	health_insurance	FR	LAMal Traitement lésions système mastication
BGE 126 V 319	health_insurance	IT	LAMal Trattamento lesioni sistema masticatorio
BGE 132 V 393	old_court_law	DE	OG seit Juli geltenden Fassung
BGE 132 V 393	old_court_law	FR	OJ teneur vigueur depuis juillet
BGE 132 V 393	old_court_law	IT	OG versione vigore luglio Cognizione

Table 15: Cross-lingual question set (page 4 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 144 V 210	misc	DE	misc Verordnung Europäischen Parlaments Rates
BGE 144 V 210	misc	FR	misc Règlement Parlement européen Conseil
BGE 144 V 210	misc	IT	misc Regolamento Parlamento europeo Consiglio
BGE 136 IV 55	criminal_code	DE	Strafgesetzbuch StGB Strafzumessung verminderter Schuldfähigkeit
BGE 136 IV 55	criminal_code	FR	CP fixation peine responsabilité restreinte
BGE 136 IV 55	criminal_code	IT	CP commisurazione pena caso scemata
BGE 142 III 413	civil_procedure	DE	Zivilprozessordnung Novenrecht Berufungsinstanz Grundsätze Berufungsverfahrens
BGE 142 III 413	civil_procedure	FR	CPC recevabilité nova appel Principes
BGE 142 III 413	civil_procedure	IT	CPC nova sede appello Principi
BGE 110 V 48	misc	DE	misc Anfechtungsgegenstand Streitgegenstand Verfügung Sachurteilsvoraussetzung
BGE 110 V 48	misc	FR	misc Objet contestation litige existence
BGE 110 V 48	misc	IT	misc Oggetto impugnato litigioso esistenza
BGE 137 III 617	civil_procedure	DE	Zivilprozessordnung Berufungsanträge Berufungseingabe Anträge enthalten
BGE 137 III 617	civil_procedure	FR	CPC conclusions appel mémoire doit
BGE 137 III 617	civil_procedure	IT	CPC conclusioni appello allegato deve
BGE 129 V 1	old_age_insurance	DE	AHVG Ende gültig gewesen Fassung
BGE 129 V 1	old_age_insurance	FR	LAVS teneur vigueur jusqu corrélation
BGE 129 V 1	old_age_insurance	IT	LAVS tenore vigente fino dicembre
BGE 126 I 97	criminal_code	DE	Strafgesetzbuch BStP Zulässigkeit Geschädigten gegen
BGE 126 I 97	criminal_code	FR	CP Recevabilité droit public interjeté
BGE 126 I 97	criminal_code	IT	CP Ammissibilità diritto pubblico inoltrato
BGE 143 III 65	civil_code	DE	Zivilgesetzbuch Anhörung Pflegeeltern Entlassung Beiständin
BGE 143 III 65	civil_code	FR	CC audition parents nourriciers libération
BGE 143 III 65	civil_code	IT	CC audizione genitori affilianti dimissione
BGE 121 V 362	unemployment_insurance	DE	Arbeitslosenversicherungsgesetz Stellung Personals öffentlicher Dienste
BGE 121 V 362	unemployment_insurance	FR	LACI Statut personnel services publics
BGE 121 V 362	unemployment_insurance	IT	LADI Statuto personale servizi pubblici
BGE 134 IV 1	criminal_code	DE	Strafgesetzbuch Bedingte teilbedingte Strafe gemäss
BGE 134 IV 1	criminal_code	FR	CP Sursis partiel exécution peine
BGE 134 IV 1	criminal_code	IT	CP Pene condizionale parziale giusta

Table 16: Cross-lingual question set (page 5 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 144 IV 345	criminal_procedure	DE	Strafprozessordnung StPO Unschuldsvermutung Beweiswürdigung Wiederholung
BGE 144 IV 345	criminal_procedure	FR	CPP présomption innocence appréciation preuves
BGE 144 IV 345	criminal_procedure	IT	CPP Cost presunzione innocenza valutazione
BGE 128 III 411	civil_code	DE	Zivilgesetzbuch Kinderunterhaltsbeitrag Untersuchungsmaxime Tragweite Diese
BGE 128 III 411	civil_code	FR	CC contribution entretien faveur enfants
BGE 128 III 411	civil_code	IT	CC contributo mantenimento prole massima
BGE 146 IV 88	criminal_procedure	DE	Strafprozessordnung Verweigerung Massnahmen Feststellung Fahruntfähigkeit
BGE 146 IV 88	criminal_procedure	FR	CPP lien refus mesures constatation
BGE 146 IV 88	criminal_procedure	IT	CPP unitamente LCStr rifiuto sottoporsi
BGE 144 III 349	civil_procedure	DE	Zivilprozessordnung Zulässigkeit Noven Berufungsverfahren uneingeschränkte
BGE 144 III 349	civil_procedure	FR	CPC recevabilité nova appel maxime
BGE 144 III 349	civil_procedure	IT	CPC ammissibilità nova appello principio
BGE 129 I 129	old_court_law	DE	OG Anspruch unentgeltlichen Rechtsbeistand anfechtbarer
BGE 129 I 129	old_court_law	FR	OJ droit être assisté gratuitement
BGE 129 I 129	old_court_law	IT	OG Cost diritto avvocato ufficio
BGE 142 I 135	foreigners_law	DE	AuG Zulässigkeit öffentlich-rechtlichen Angelegenheiten gegen
BGE 142 I 135	foreigners_law	FR	AuG LEtr admissibilité matière droit
BGE 142 I 135	foreigners_law	IT	AuG Cost LStr ammissibilità materia
BGE 138 I 232	contract_tort	DE	Obligationenrecht Personalreglement Öffentlichen Verkehrsbetriebe Genf
BGE 138 I 232	contract_tort	FR	CO Statut personnel Transports publics
BGE 138 I 232	contract_tort	IT	CO Cost segg Statuto personale
BGE 130 II 281	echr	DE	EMRK Familiennachzug gefestigtes Anwesenheitsrecht kombinierten
BGE 130 II 281	echr	FR	CEDH regroupement familial droit présence
BGE 130 II 281	echr	IT	CEDU ricongiungimento familiare diritto residenza
BGE 142 IV 137	misc	DE	misc qualifiziert grobe Verkehrsregelverletzung besonders
BGE 142 IV 137	misc	FR	misc violation grave qualifiée règles
BGE 142 IV 137	misc	IT	misc LCStr grave infrazione qualificata
BGE 123 V 150	unemployment_insurance	DE	Arbeitslosenversicherungsgesetz Verwaltungspraxis wonach Versicherte unwahren
BGE 123 V 150	unemployment_insurance	FR	LACI Constitue abus pouvoir appréciation
BGE 123 V 150	unemployment_insurance	IT	LADI costitutiva abuso potere apprezzamento

D MCP liveness telemetry (aggregate)

We log only request user-agent and request volume on the public MCP endpoint. We do not log query payloads, response payloads, or any client-matter facts. The aggregate counts in Table 17 cover the window 12 April–9 May 2026 (four weeks). Per the licence and privacy posture in Sec. 11, this telemetry is not a measure of correctness or professional reliance — only of liveness.

Table 17: Aggregate MCP liveness signal from tier-1 server logs. Window: 12 April–9 May 2026. We log the user-agent string and request volume only; we do not log query payloads or downstream client-matter facts. Adoption is descriptive evidence the resource is exercised by external clients; it is not an evaluation claim about quality, correctness, or professional reliance.

Client category (identified by user-agent)	Requests
Agentic-coding CLI integrations	~2,300
MCP-front-end integrations from major LLM vendors	~2,000
IDE-based integrations	~2,600
Workflow-platform integrations (anonymised)	~4,100
Generic MCP-discovery probes (77 distinct IPs)	varied
Search-engine and LLM-crawler indexing	—